MATHS MANTHAN

• TUITION CLASSES •

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Subject: Mathematics | **WEEKLY CHAPTER TEST – 04 (POLYNOMIALS)** | **Time:** 45 Minutes

Class: Class X (CBSE) | **Chapter 2: Polynomials** | **MM: 20**

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CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, zero skipped steps, and final boxed answers with proper units.
2. In alternative (**OR**) sub-questions, complete step-by-step solutions and equal step marking apply to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Roots rejected by physical domain/restrictions are explicitly stated.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice Questions

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Q1. If one of the zeroes of a quadratic polynomial of the form $x^2 + ax + b$ is the negative of the other, then it: [1]

(A) no linear term, constant term is negative **(B)** no linear term, constant term is positive

(C) can have linear term, constant term negative **(D)** can have linear term, constant term positive

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Let the zeroes of $x^2 + ax + b$ be α and $-\alpha$ ($\alpha \neq 0$ for distinct opposite zeroes). We know Vieta's relations:

Formula Used: Sum of zeroes $= -a$, Product of zeroes $= b$

Here $\alpha + (-\alpha) = 0 \implies -a = 0 \implies a = 0$ (no linear term ax), and $\alpha(-\alpha) = -\alpha^2 = b < 0$ (constant term is negative). Thus $\boxed{\text{(A)}}$ is correct. [1M]

Q2. If the zeroes of the quadratic polynomial $ax^2 + bx + c$, $c \neq 0$ are equal, then: [1]

(A) c and a have opposite signs **(B)** c and b have opposite signs

(C) c and a have the same sign **(D)** c and b have the same sign

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C)

We know that a quadratic polynomial $ax^2 + bx + c$ ($a \neq 0, c \neq 0$) has equal real zeroes if and only if:

Formula Used: $D = b^2 - 4ac = 0 \implies b^2 = 4ac$

Since $b^2 \geq 0$ for all real b and $ac \neq 0$ (as $a \neq 0, c \neq 0$), we must have $4ac > 0 \implies ac > 0$. Thus c and a are either both positive or both negative, i.e., $\boxed{\text{(C) same sign}}$. [1M]

Q3. If the sum of the zeroes of the quadratic polynomial $f(t) = kt^2 + 2t + 3k$ is equal to their product, then $k =$: [1]

(A) $\frac{2}{3}$ (B) $-\frac{2}{3}$ (C) $\frac{3}{2}$ (D) $-\frac{3}{2}$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B)

Comparing $f(t) = kt^2 + 2t + 3k$ $(k \neq 0)$ with $at^2 + bt + c$, we have $a = k$, $b = 2$, $c = 3k$. We know:

Formula Used: $\alpha + \beta = -\frac{b}{a}, \qquad \alpha\beta = \frac{c}{a}$

Setting $\alpha + \beta = \alpha\beta$:

$$-\frac{2}{k} = \frac{3k}{k} \implies -\frac{2}{k} = 3 \implies 3k = -2 \implies \boxed{k = -\frac{2}{3}}. \qquad [1M]$$

Q4. If the product of the zeroes of the quadratic polynomial $f(x) = x^2 - 4x + k$ is 3, then the value of k is: [1]

(A) 3 (B) -3 (C) 4 (D) -4

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Comparing $f(x) = x^2 - 4x + k$ with $ax^2 + bx + c$, we have $a = 1$ and $c = k$. We know:

Formula Used: Product of zeroes $= \alpha\beta = \frac{c}{a}$

Since the product of zeroes is 3:

$$\frac{k}{1} = 3 \implies \boxed{k = 3}. \qquad [1M]$$

Q5. Assertion-Reason: [1]

Assertion (A): A quadratic polynomial having $2 + \sqrt{3}$ and $2 - \sqrt{3}$ as its zeroes is $x^2 - 4x + 1$.

Reason (R): A quadratic polynomial with zeroes α and β is given by $k[x^2 - (\alpha + \beta)x + \alpha\beta]$, where $k \neq 0$.

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) Both A and R are true and R is the correct explanation of A.

Let $\alpha = 2 + \sqrt{3}$ and $\beta = 2 - \sqrt{3}$. We know:

Formula Used: $p(x) = k[x^2 - (\alpha + \beta)x + \alpha\beta]$

Computing $S = (2 + \sqrt{3}) + (2 - \sqrt{3}) = 4$ and $P = (2 + \sqrt{3})(2 - \sqrt{3}) = 4 - 3 = 1$. Setting $k = 1$ gives $p(x) = x^2 - 4x + 1$. Thus both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**. [1M]

Q6. Find the zeroes of $f(x) = 4\sqrt{3}\,x^2 + 5x - 2\sqrt{3}$ and verify the relationship between the zeroes and coefficients. **[2]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 2 Marks]

Comparing $f(x)$ with $ax^2 + bx + c$, we have $a = 4\sqrt{3}$, $b = 5$, $c = -2\sqrt{3}$. We know:

Formula Used: $\alpha + \beta = -\dfrac{b}{a}, \qquad \alpha\beta = \dfrac{c}{a}$

Step 1 – Factorisation: Product $a \cdot c = (4\sqrt{3})(-2\sqrt{3}) = -24$; splitting $+5 = +8 - 3$:

$$\begin{aligned} f(x) &= 4\sqrt{3}\,x^2 + 8x - 3x - 2\sqrt{3} \\ &= 4x(\sqrt{3}\,x + 2) - \sqrt{3}(\sqrt{3}\,x + 2) = (4x - \sqrt{3})(\sqrt{3}\,x + 2). \end{aligned}$$

Setting $f(x) = 0$ gives the zeroes $\alpha = \boxed{\dfrac{\sqrt{3}}{4}}$ and $\beta = \boxed{-\dfrac{2}{\sqrt{3}}}$. **[1M]**

Step 2 – Verification:

$$\alpha + \beta = \frac{\sqrt{3}}{4} - \frac{2}{\sqrt{3}} = \frac{3-8}{4\sqrt{3}} = -\frac{5}{4\sqrt{3}}, \qquad -\frac{b}{a} = -\frac{5}{4\sqrt{3}}, \qquad \textbf{[0.5M]}$$

$$\alpha\beta = \left(\frac{\sqrt{3}}{4}\right)\left(-\frac{2}{\sqrt{3}}\right) = -\frac{1}{2}, \qquad \frac{c}{a} = \frac{-2\sqrt{3}}{4\sqrt{3}} = -\frac{1}{2}. \qquad \textbf{[0.5M]}$$

Q7. If α and β are the zeroes of the polynomial $f(x) = 2x^2 + 5x + k$ such that $\alpha^2 + \beta^2 + \alpha\beta = \dfrac{21}{4}$, find the value of k. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

For $f(x) = 2x^2 + 5x + k$, we have $a = 2$, $b = 5$, $c = k$. We know Vieta's formulas and the square identity:

Formula Used: $\alpha + \beta = -\dfrac{b}{a}, \quad \alpha\beta = \dfrac{c}{a}, \qquad \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

Thus:

$$\alpha + \beta = -\frac{5}{2}, \qquad \alpha\beta = \frac{k}{2}. \qquad \textbf{[1M]}$$

Rewriting the given condition in terms of $(\alpha + \beta)$ and $\alpha\beta$:

$$\alpha^2 + \beta^2 + \alpha\beta = [(\alpha + \beta)^2 - 2\alpha\beta] + \alpha\beta = (\alpha + \beta)^2 - \alpha\beta. \qquad \textbf{[1M]}$$

Substituting the given value $\dfrac{21}{4}$:

$$\left(-\frac{5}{2}\right)^2 - \frac{k}{2} = \frac{21}{4} \implies \frac{25}{4} - \frac{k}{2} = \frac{21}{4} \implies \frac{k}{2} = \frac{25-21}{4} = 1 \implies \boxed{k = 2}. \qquad \textbf{[1M]}$$

Q8. If the sum of squares of zeroes of the quadratic polynomial $f(x) = x^2 - 8x + k$ is equal to 40, find the value of k. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Let α and β be the zeroes of $f(x) = x^2 - 8x + k$ ($a = 1$, $b = -8$, $c = k$). We know:

Formula Used: $\alpha + \beta = -\dfrac{b}{a}, \quad \alpha\beta = \dfrac{c}{a}, \quad \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

Thus:

$$\alpha + \beta = -\frac{(-8)}{1} = 8, \qquad \alpha\beta = \frac{k}{1} = k. \qquad \text{[1M]}$$

Given $\alpha^2 + \beta^2 = 40$:

$$(\alpha + \beta)^2 - 2\alpha\beta = 40 \qquad \text{[1M]}$$

$$8^2 - 2k = 40 \implies 64 - 2k = 40 \implies 2k = 24 \implies \boxed{k = 12}. \qquad \text{[1M]}$$

Q9. If α, β are the zeroes of $f(x) = x^2 - p(x + 1) - c$, prove that: $(\alpha + 1)(\beta + 1) = 1 - c$. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

First expand $f(x)$ into standard quadratic form $Ax^2 + Bx + C$:

$$f(x) = x^2 - px - p - c = x^2 - px - (p + c), \quad \text{so } A = 1,\ B = -p,\ C = -(p + c). \qquad \text{[1M]}$$

We know Vieta's sum and product of zeroes relations:

Formula Used: $\alpha + \beta = -\dfrac{B}{A}, \qquad \alpha\beta = \dfrac{C}{A}$

Therefore:

$$\alpha + \beta = p, \qquad \alpha\beta = -(p + c). \qquad \text{[1M]}$$

Expanding the LHS and substituting these values:

$$\text{LHS} = (\alpha + 1)(\beta + 1) = \alpha\beta + \alpha + \beta + 1$$

$$= -(p + c) + p + 1 = -p - c + p + 1 = \boxed{1 - c = \text{RHS.}} \qquad \text{[1M]}$$

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Q10. Case Study: Suspension Bridge Cable Arch

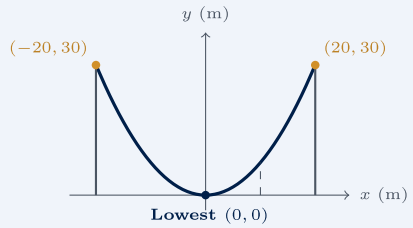
A suspension bridge cable hangs in a symmetrical parabola $y = ax^2 + bx + c$, anchored at towers $(-20, 30)$ and $(20, 30)$ with its lowest point at the origin $(0, 0)$. [4]

(i) Using the coordinates of the lowest point and symmetry about the y-axis, determine b and c. **[1]**

(ii) Find the value of coefficient a and write the quadratic equation of the suspension cable curve. **[1]**

(iii) **(a)** Find the height of the suspension cable above the road deck at 10 m from the centre. **OR** **(b)** Determine the length of the vertical suspender rod located at $x = 15$ m. **[2]**

▶ CASE-BASED SOLUTION & MARKING SCHEME: [Total: 4 Marks]



Arch Symmetry: With the vertex anchored at $(0, 0)$ and symmetric tower ends at $(-20, 30)$ and $(20, 30)$, the parabolic curve is an even function $y = ax^2$ ($b = c = 0$).

Formula Used: $y = ax^2 + bx + c$

(i) Finding b and c: Since the cable passes through $(0, 0)$, substituting $x = 0, y = 0$ gives $c = \boxed{0}$. **[0.5M]**

By symmetry, the curve passes through both $(-20, 30)$ and $(20, 30)$, so $y(-20) = y(20)$:

$$400a - 20b = 400a + 20b \implies 40b = 0 \implies \boxed{b = 0}. \quad \textbf{[0.5M]}$$

(ii) Finding a & equation: Substituting $b = 0, c = 0$ and $(20, 30)$ into $y = ax^2$:

$$30 = a(20)^2 \implies 400a = 30 \implies a = \frac{30}{400} = \boxed{\frac{3}{40}} \implies \boxed{y = \frac{3}{40}x^2}. \quad \textbf{[1M]}$$

(iii)(a) Height at $x = 10$ m: Substituting $x = 10$ into $y = \frac{3}{40}x^2$:

$$y = \frac{3}{40}(10)^2 = \frac{300}{40} = \frac{15}{2} = \boxed{7.5 \text{ m}}. \quad \textbf{[2M]}$$

OR (Alternative Part (b) – Rod length at $x = 15$ m): Since the deck is at $y = 0$, the suspender rod length equals the cable height at $x = 15$ m:

$$y = \frac{3}{40}(15)^2 = \frac{3 \times 225}{40} = \frac{135}{8} = \boxed{16.875 \text{ m}}. \quad \textbf{[2M]}$$

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